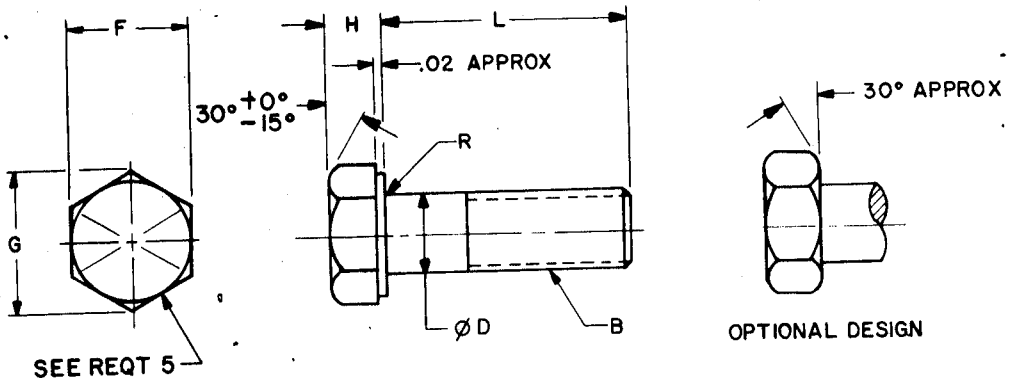


TABLE I. DIMENSIONS AND PART NUMBERS.

NOMINAL SIZE		.250	.375
B - THREADS PER INCH UNC-2A		20	16
Ø D - BODY DIAMETER	MAX	.2500	.3750
	MIN	.2450	.3690
F - WIDTH ACROSS FLATS	MAX	.4375	.5625
	MIN	.4280	.5510
G - WIDTH ACROSS CORNERS	MAX	.5050	.6500
	MIN	.4880	.6280
H - HEAD HEIGHT	MAX	.1630	.2430
	MIN	.1500	.2260
R - RADIUS	MAX	.0250	.0250
	MIN	.0150	.0150
1/ MINIMUM TENSILE LOAD - LBS		4,750	11,600
L - LENGTH		DASH NO	DASH NO
.625 +.000 −.031		1	3
1.000 +.000 −.062		2	4

USER ACTIVITIES:
ARMY - MI
NAVY - MC, OS

REVIEWER ACTIVITIES:
ARMY - AV, EA
AIR FORCE
DLA - IS
NSA - NS

This military standard is approved for use by all Departments and Agencies of the Department of Defense. Activation for all new engineering and design applications and for repetitive use shall be made from this document when applicable.

APPROVED 1 AUG 84
REVISED

P.A.	AR	INTERNATIONAL INTEREST	TITLE	MILITARY STANDARD
Other Cust	SH 99		SCREW, CAP HEXAGON HEAD (FINISHED HEXAGON BOLT), ALLOY STEEL, GRADE 8	MS51488
PROCUREMENT SPECIFICATION: FF-S-85		SUPERSEDES:		PAGE 1 OF 2

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5305

USER ACTIVITIES:
ARMY-MI
NAVY-MC,OS

REVIEWER ACTIVITIES:
ARMY-AV,EA
AIR FORCE-82
DLA-IS
NSA-NS

REQUIREMENTS:

1. MATERIAL. ALLOY STEEL, GRADE 8, IN ACCORDANCE WITH PROCUREMENT SPECIFICATION.

2. FINISH. ZINC COATING IN ACCORDANCE WITH ASTM B633-78, Fe/Zn_5 (5 μm), TYPE II.

3. THREADS. THE THREADS SHALL BE IN ACCORDANCE WITH FED-STD-H28/2. ACCEPTABILITY OF SCREW THREADS SHALL BE IN ACCORDANCE WITH FED-STD-H28/20, SYSTEM 21.

4. THREAD LENGTH. MINIMUM THREAD LENGTH SHALL BE TWICE THE DIAMETER PLUS .250 INCH. THE TOLERANCE SHALL BE PLUS 1.88 INCH OR 2-1/2 THREADS WHICHEVER IS GREATER. ON SCREWS THAT ARE TOO SHORT FOR MINIMUM THREAD LENGTHS, THE DISTANCE FROM THE BEARING SURFACE OF THE HEAD TO THE FIRST COMPLETE THREAD SHALL NOT EXCEED THE LENGTH OF 2-1/2 THREADS FOR SIZES UP TO AND INCLUDING 1.000 INCH.

5. HEAD MARKING. HEADS SHALL BE MARKED IN ACCORDANCE WITH MIL-HDBK-131. RAISED MARKINGS ARE PREFERRED, STAMPINGS ARE ACCEPTABLE.

6. BASED ON MINIMUM ULTIMATE TENSILE STRENGTH OF 150,000 PSI FOR DIAMETERS THRU 1.500 INCHES. TENSILE LOADS ARE CALCULATED BY THE STRESS AREAS INDICATED IN FED-STD-H28/2.

7. PART NUMBERS. THE BASIC MS PART NUMBER IS FOLLOWED BY A DASH NUMBER TAKEN FROM TABLE I.

   EXAMPLE: MS51488-1 - SCREW, CAP, HEXAGON HEAD, ALLOY STEEL, GRADE 8, .250-20 UNC-2A, LENGTH .625, ZINC COATED

NOTES:

1. ALL DIMENSIONS ARE IN INCHES.

2. REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATIONS FOR BIDS OR REQUEST FOR PROPOSAL, EXCEPT THAT REFERENCED ADOPTED INDUSTRY DOCUMENTS SHALL GIVE THE DATE OF THE ISSUE ADOPTED.

3. FOR DESIGN FEATURE PURPOSES, THIS STANDARD TAKES PRECEDENCE OVER PROCUREMENT DOCUMENTS REFERENCED HEREIN.

APPROVED 1 AUG 84 REVISED

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P.A. AR / Other Cust SH 99	INTERNATIONAL INTEREST	TITLE: SCREW, CAP HEXAGON HEAD (FINISHED HEXAGON BOLT),	MILITARY STANDARD MS51488
PROCUREMENT SPECIFICATION: FF-S-85	SUPERSEDES:		PAGE 2 OF 2

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